PYTHON FOR DATA SCIENCE · LESSON 17 OF 31

pandas II: Data Cleaning: Missing Values, Types, Duplicates, Text and Outliers

Turn messy data into analysis-ready data. Detect and treat missing values, fix data types, remove duplicates, standardise text, parse dates, handle outliers with IQR and caps, and document every step in a reusable cleaning function.

Python class 13 | pandas 2.3 / 3.0 | Intermediate | 120 min

By the end of this lesson you can

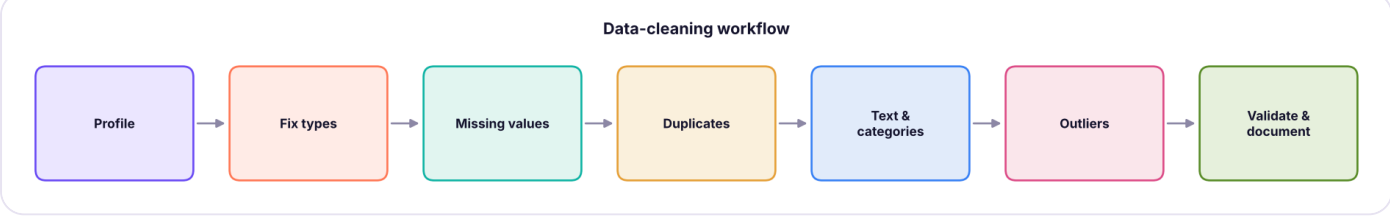
- Profile a dataset for missing values, wrong types, duplicates and inconsistent categories
- Choose and apply missing-value strategies (drop, fill with statistic, group-wise fill, flag)
- Convert types safely with to_numeric, to_datetime and astype, including categories
- Standardise text columns and map inconsistent labels
- Detect outliers with the IQR rule and z-scores and decide how to treat them

Datasets used

`house_prices.csv` `hr_employees.csv`

Why cleaning matters

Analysts often say 60–80% of project time goes into preparing data. Wrong types, gaps, duplicates and typos silently break totals, joins and models: **garbage in, garbage out**.



A repeatable cleaning workflow.

We will clean a deliberately messy extract:

Python

```
import pandas as pd, numpy as np
pd.set_option("display.max_columns", 20); pd.set_option("display.width", 160)
raw = pd.DataFrame({
    "order_id":  ["A1", "A2", "A3", "A3", "A4", "A5", "A6", "A7", "A8"],
    "city":      [" pune", "Pune ", "DELHI", "DELHI", "New Delhi", "kochi", None,
"Bengaluru", "Bangalore"],
    "amount":    ["1,200", "950", "₹3,400", "₹3,400", "abc", "780", "15000", "2,050",
None],
    "order_date":["2025-01-05", "05/01/2025", "2025-01-07", "2025-01-07", "2025-02-30",
"2025-01-09", "2025-01-10", "2025-01-11", "2025-01-12"],
    "qty":       [2, 1, 3, 3, None, 1, 50, 2, 1],
})
print(raw)
print(raw.dtypes.to_dict())
```

OUTPUT

```
  order_id       city  amount  order_date   qty
0       A1       pune   1,200  2025-01-05   2.0
1       A2      Pune       950  05/01/2025   1.0
2       A3      DELHI  ₹3,400  2025-01-07   3.0
3       A3      DELHI  ₹3,400  2025-01-07   3.0
4       A4  New Delhi     abc  2025-02-30   NaN
5       A5      kochi     780  2025-01-09   1.0
6       A6        NaN   15000  2025-01-10  50.0
7       A7  Bengaluru   2,050  2025-01-11   2.0
8       A8  Bangalore     NaN  2025-01-12   1.0
{'order_id': <StringDtype(na_value=nan)>, 'city': <StringDtype(na_value=nan)>,
'amount': <StringDtype(na_value=nan)>, 'order_date': <StringDtype(na_value=nan)>,
'qty': dtype('float64')}
```

Step 1: Profile

Python

```
def profile(df):
    return pd.DataFrame({"dtype": df.dtypes.astype(str), "missing": df.isna().sum(),
"missing_%": (100 * df.isna().mean()).round(1),
                         "unique": df.nunique(), "example": df.iloc[0]})
print(profile(raw))
print("duplicate rows:", raw.duplicated().sum())
```

OUTPUT

```
              dtype  missing  missing_%  unique     example
order_id        str        0        0.0       8          A1
```

```
city            str        1       11.1       7        pune
amount          str        1       11.1       7       1,200
order_date      str        0        0.0       8  2025-01-05
qty         float64        1       11.1       4         2.0
duplicate rows: 1
```

Step 2: Fix types

Python

```python
df = raw.drop_duplicates().copy()                                    # exact
duplicates first
df["amount"] = pd.to_numeric(df["amount"].str.replace(r"[₹,]", "", regex=True),
errors="coerce")   # bad values → NaN
df["order_date"] = pd.to_datetime(df["order_date"], format="%Y-%m-%d", errors="coerce")
# strict format: others → NaT
df["qty"] = df["qty"].astype("Int64")                                 # nullable
integer keeps NaN
print(df[["order_id", "amount", "order_date", "qty"]])
print(df.dtypes.to_dict())
```

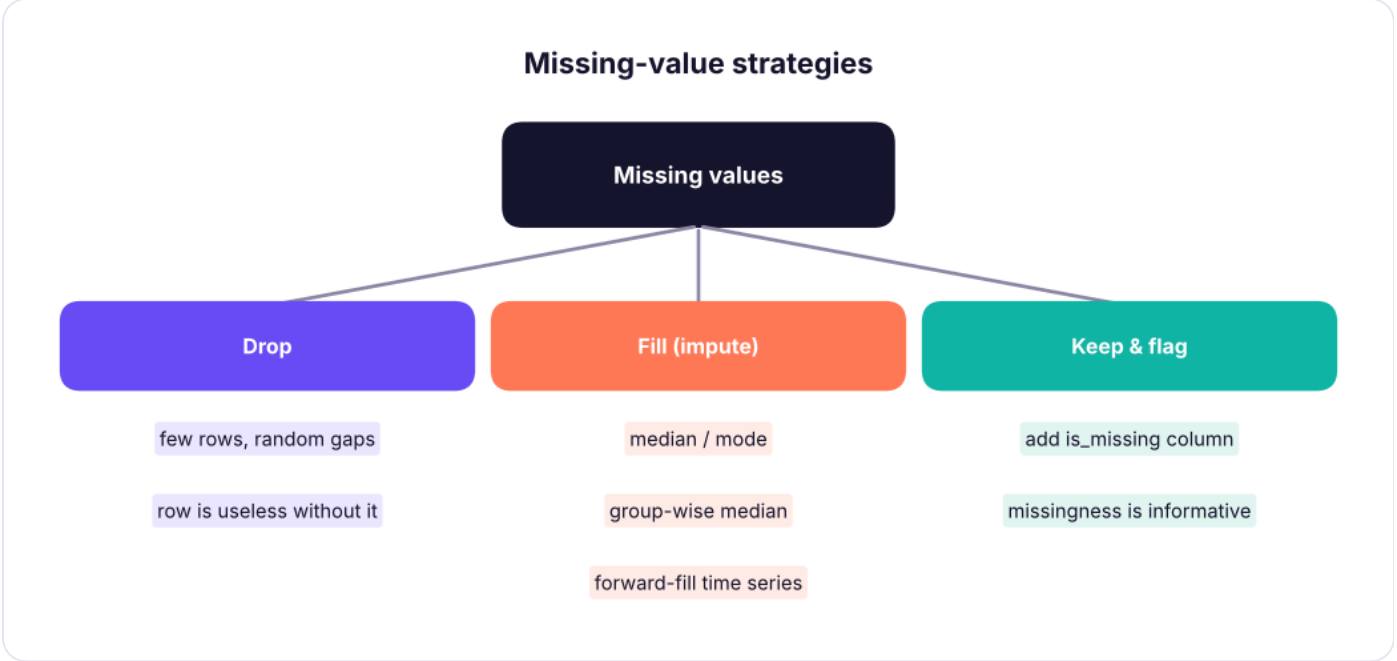
OUTPUT

```
  order_id   amount order_date   qty
0       A1   1200.0 2025-01-05     2
1       A2    950.0        NaT     1
2       A3   3400.0 2025-01-07     3
4       A4      NaN        NaT  <NA>
5       A5    780.0 2025-01-09     1
6       A6  15000.0 2025-01-10    50
7       A7   2050.0 2025-01-11     2
8       A8      NaN 2025-01-12     1
{'order_id': <StringDtype(na_value=nan)>, 'city': <StringDtype(na_value=nan)>,
'amount': dtype('float64'), 'order_date': dtype('<M8[us]'), 'qty': Int64Dtype()}
```

`errors="coerce"` turns unparseable values into `NaN` / `NaT` so you can count and inspect them instead of crashing. Always check how many values were coerced.

Conversion	Use
`pd.to_numeric(s, errors="coerce")`	Text with numbers
`pd.to_datetime(s, format=..., dayfirst=True)`	Dates (give the format when known)
`s.astype("category")`	Repeated labels: less memory, ordered categories
`s.astype("Int64")`, `"boolean"`, `"string"`	Nullable types that allow missing values

Step 3: Missing values



Choosing a strategy for missing values.

Python

```
print(df.isna().sum().to_dict())
df["amount_missing"] = df["amount"].isna()                    # flag before
filling
df["amount"] = df["amount"].fillna(df["amount"].median())
df["qty"] = df["qty"].fillna(1)
df = df.dropna(subset=["order_date"])                         # a row without a
valid date is unusable here
print(df[["order_id", "amount", "amount_missing", "qty", "order_date"]])
```

OUTPUT

```
{'order_id': 0, 'city': 1, 'amount': 2, 'order_date': 2, 'qty': 1}
  order_id   amount  amount_missing  qty order_date
0       A1   1200.0           False    2 2025-01-05
2       A3   3400.0           False    3 2025-01-07
```

```
5       A5    780.0          False    1 2025-01-09
6       A6  15000.0          False   50 2025-01-10
7       A7   2050.0          False    2 2025-01-11
8       A8   1625.0           True    1 2025-01-12
```

Real data: `house_prices.csv` has gaps in `area_sqft`. A **group-wise** fill uses similar houses (same number of bedrooms) instead of one global value:

Python

```
h = pd.read_csv("house_prices.csv")
print("missing area:", h["area_sqft"].isna().sum())
by_bed = h.groupby("bedrooms")["area_sqft"].median().round(0)
print(by_bed.to_dict())
h["area_filled"] = h["area_sqft"].fillna(h.groupby("bedrooms")
["area_sqft"].transform("median"))
print("after group-wise fill:", h["area_filled"].isna().sum(), "| global median would
give", h["area_sqft"].median())
```

OUTPUT

```
missing area: 25
{1: 760.0, 2: 1115.0, 3: 1430.0, 4: 1715.0, 5: 1955.0}
after group-wise fill: 0 | global median would give 1240.0
```

NOTE **Missing-data types**

MCAR (completely at random) → dropping is safe. **MAR** (depends on other columns) → impute using those columns. **MNAR** (depends on the missing value itself, e.g. high earners hide income) → flag it; imputation can bias results. Never fill target variables for modelling.

Step 4: Duplicates

Python

```
orders = pd.DataFrame({"order_id": ["B1", "B2", "B2", "B3"], "status": ["new", "new",
"shipped", "new"], "updated": pd.to_datetime(["2025-01-01", "2025-01-01", "2025-01-03",
"2025-01-02"])})
print(orders.duplicated().sum(), orders.duplicated(subset="order_id").sum())
latest = orders.sort_values("updated").drop_duplicates(subset="order_id", keep="last")   #
keep the latest version per key
print(latest)
```

OUTPUT

```
0 1
  order_id   status    updated
0       B1      new 2025-01-01
3       B3      new 2025-01-02
2       B2  shipped 2025-01-03
```

Decide which columns define "the same record"; exact duplicates and key duplicates are different problems.

Step 5: Text and categories

Python

```python
df["city"] = df["city"].str.strip().str.title()
fix = {"New Delhi": "Delhi", "Bangalore": "Bengaluru"}
df["city"] = df["city"].replace(fix).fillna("Unknown")
print(df["city"].value_counts().to_dict())
print(pd.Series(["  Data  Science ", "MACHINE learning"]).str.split().str.join("
").str.capitalize().tolist())
```

OUTPUT

```
{'Bengaluru': 2, 'Pune': 1, 'Delhi': 1, 'Kochi': 1, 'Unknown': 1}
['Data science', 'Machine learning']
```

Useful `.str` methods: `strip`, `lower/upper/title`, `replace`, `contains`, `startswith`, `extract`, `split`, `len`, `zfill`, `slice`. For fuzzy spellings, libraries such as **rapidfuzz** score string similarity.

Step 6: Outliers

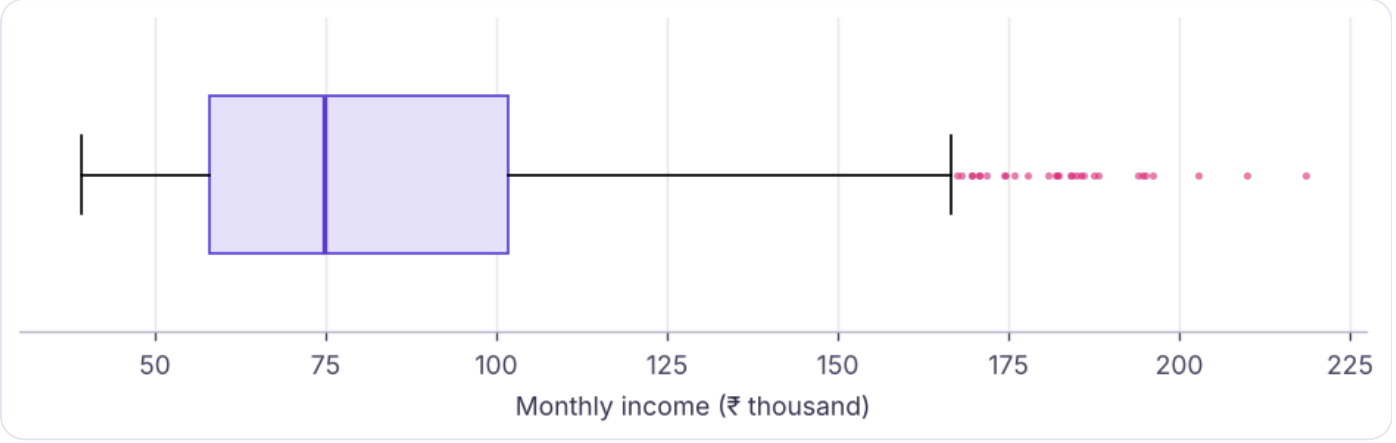
An **outlier** is far from most values. It may be an error (qty = 50 typed instead of 5) or real (one bulk order). Detect first, then decide.

Python

```python
hr = pd.read_csv("hr_employees.csv")
inc = hr["monthly_income"]
q1, q3 = inc.quantile([0.25, 0.75]); iqr = q3 - q1
lo, hi = q1 - 1.5 * iqr, q3 + 1.5 * iqr
print(f"Q1 {q1:,.0f} | Q3 {q3:,.0f} | IQR fences {lo:,.0f} to {hi:,.0f}")
out = hr[(inc < lo) | (inc > hi)]
print(len(out), "IQR outliers;", out["job_title"].value_counts().head(3).to_dict())
z = (inc - inc.mean()) / inc.std()
print((z.abs() > 3).sum(), "rows with |z| > 3")
capped = inc.clip(lower=inc.quantile(0.01), upper=inc.quantile(0.99))    # winsorise at 1st/99th percentile
print(f"max before {inc.max():,.0f} after capping {capped.max():,.0f}")
```

OUTPUT

```
Q1 58,062 | Q3 101,833 | IQR fences -7,594 to 167,489
29 IQR outliers; {'Sales Manager': 6, 'Finance Manager': 5, 'Operations Manager': 5}
19 rows with |z| > 3
max before 218,417 after capping 182,086
```



Monthly income: the box plot marks values beyond the IQR fences as points.

Senior roles earn more, so high incomes here are **real**, not errors. Options: keep them, analyse by job level, use medians, log-transform, or cap for models that are sensitive to extremes. Delete only values that are clearly impossible (negative age, qty 50 in a 1–5 item shop).

A reusable cleaning function

Python

```
def clean_orders(raw: pd.DataFrame) -> pd.DataFrame:
    """Return cleaned orders and print a short audit log."""
    df = raw.drop_duplicates().copy(); log = {"rows_in": len(raw), "dupes_removed":
len(raw) - len(df)}
    df["amount"] = pd.to_numeric(df["amount"].str.replace(r"[₹,]", "", regex=True),
errors="coerce")
    df["order_date"] = pd.to_datetime(df["order_date"], format="%Y-%m-%d", errors="coerce")
    log["bad_dates_dropped"] = int(df["order_date"].isna().sum()); df = df.dropna(subset=
["order_date"])
    log["amount_imputed"] = int(df["amount"].isna().sum()); df["amount"] =
df["amount"].fillna(df["amount"].median())
    df["city"] = df["city"].str.strip().str.title().replace({"New Delhi": "Delhi",
"Bangalore": "Bengaluru"}).fillna("Unknown")
    df["qty"] = df["qty"].fillna(1).clip(upper=10).astype(int); log["rows_out"] = len(df)
    print(log); return df
clean = clean_orders(raw)
assert clean["amount"].notna().all() and clean["order_id"].is_unique
print(clean)
```

OUTPUT

```
{'rows_in': 9, 'dupes_removed': 1, 'bad_dates_dropped': 2, 'amount_imputed': 1,
'rows_out': 6}
  order_id       city   amount order_date  qty
0       A1       Pune   1200.0 2025-01-05    2
2       A3      Delhi   3400.0 2025-01-07    3
5       A5      Kochi    780.0 2025-01-09    1
6       A6    Unknown  15000.0 2025-01-10   10
7       A7  Bengaluru   2050.0 2025-01-11    2
8       A8  Bengaluru   2050.0 2025-01-12    1
```

Summary

KEY TAKEAWAYS

- Profile first: dtypes, missing counts, uniques, duplicates, odd categories.
- Convert with `to_numeric` / `to_datetime(errors="coerce")` and count what was coerced; use nullable and category dtypes.
- Missing values: drop, fill (median, mode, group-wise `transform`), forward-fill, or flag; think about why they are missing.
- Define duplicates by key; `sort_values + drop_duplicates(keep="last")` keeps the latest record.
- Standardise text with `.str` and mapping dictionaries; detect outliers with IQR or z-scores and treat them deliberately.
- Wrap cleaning in a function with an audit log and assertions so it is repeatable.

Quick quiz

6 questions · tap an answer to check it

1. What does `pd.to_numeric(s, errors="coerce")` do with "abc"?

A Raises an error

B Converts it to NaN

C Keeps it as text

D Converts it to 0

Answer: B. You can then count and inspect the NaNs.

2. Which fill keeps group differences?

A `fillna(df.col.mean())`

B `fillna(df.groupby("g")["col"].transform("median"))`

C `dropna()`

D `fillna(0)`

Answer: B. `transform` returns a value per row from its group.

3. What are the IQR outlier fences?

A mean ± 2 sd

B Q1 − 1.5×IQR and Q3 + 1.5×IQR

C min and max

D median ± IQR

Answer: B. The IQR is Q3 − Q1.

4. How do you keep the latest record per order_id?

A `drop_duplicates()`

B Sort by update time, then `drop_duplicates(subset="order_id", keep="last")`

C `groupby().first()` without sorting

D `unique()`

Answer: B. Sorting decides which copy is "last".

5. Why add an `is_missing` flag before filling?

A Required by pandas

B The fact that a value was missing can itself be informative

C It speeds up filling

D To delete the column later

Answer: B. Models and analysts can use the flag.

6. Which dtype allows integers with missing values?

A int64

B Int64 (nullable)

C float32 only

D object

Answer: B. The capital-I `Int64` is pandas' nullable integer.

EXERCISE

Cleaning practice

Using `hr_employees.csv`: (1) profile every column. (2) Convert `hire_date` to datetime and compute tenure in years. (3) Check whether `years_at_company` agrees with the hire date. (4) Make `education` an ordered category. (5) Find income outliers per department using the IQR rule within each department.

Show solution (worked answer)

Python

```python
hr = pd.read_csv("hr_employees.csv")
print(profile(hr))
hr["hire_date"] = pd.to_datetime(hr["hire_date"])
hr["tenure"] = ((pd.Timestamp("2025-12-31") - hr.hire_date).dt.days / 365.25).round(1)
print((hr["tenure"].round() - hr["years_at_company"]).abs().describe())
print(hr["education"].unique())
def iqr_flag(x):
    q1, q3 = x.quantile([.25, .75]); i = q3 - q1
    return (x < q1 - 1.5 * i) | (x > q3 + 1.5 * i)
hr["income_outlier"] = hr.groupby("department")["monthly_income"].transform(iqr_flag)
print(hr.groupby("department")["income_outlier"].sum())
```

INTERVIEW CORNER

Questions you may be asked

Q: How do you handle missing values? First find out why they are missing and how many. Then drop (few, random), impute (median/mode, group-wise, model-based, forward-fill for time series) or flag. Impute after the train/test split in ML to avoid leakage.

Q: How do you treat outliers? Check whether they are errors or real. Fix or remove errors; for real extremes use robust statistics (median), transformations (log), capping, or models robust to outliers.

Q: What checks do you run before trusting a dataset? Row counts vs source, uniqueness of keys, missing rates, value ranges, valid categories, date ranges, and totals reconciled to a trusted report.

Practise more online

Free, trusted places to test yourself on this topic. Links open the original site in a new tab.